**CSOPESY Semi-Major Output 2**

Prepared By: Gregory Cu
Created By: Neil Patrick Del Gallego, PhD

By group

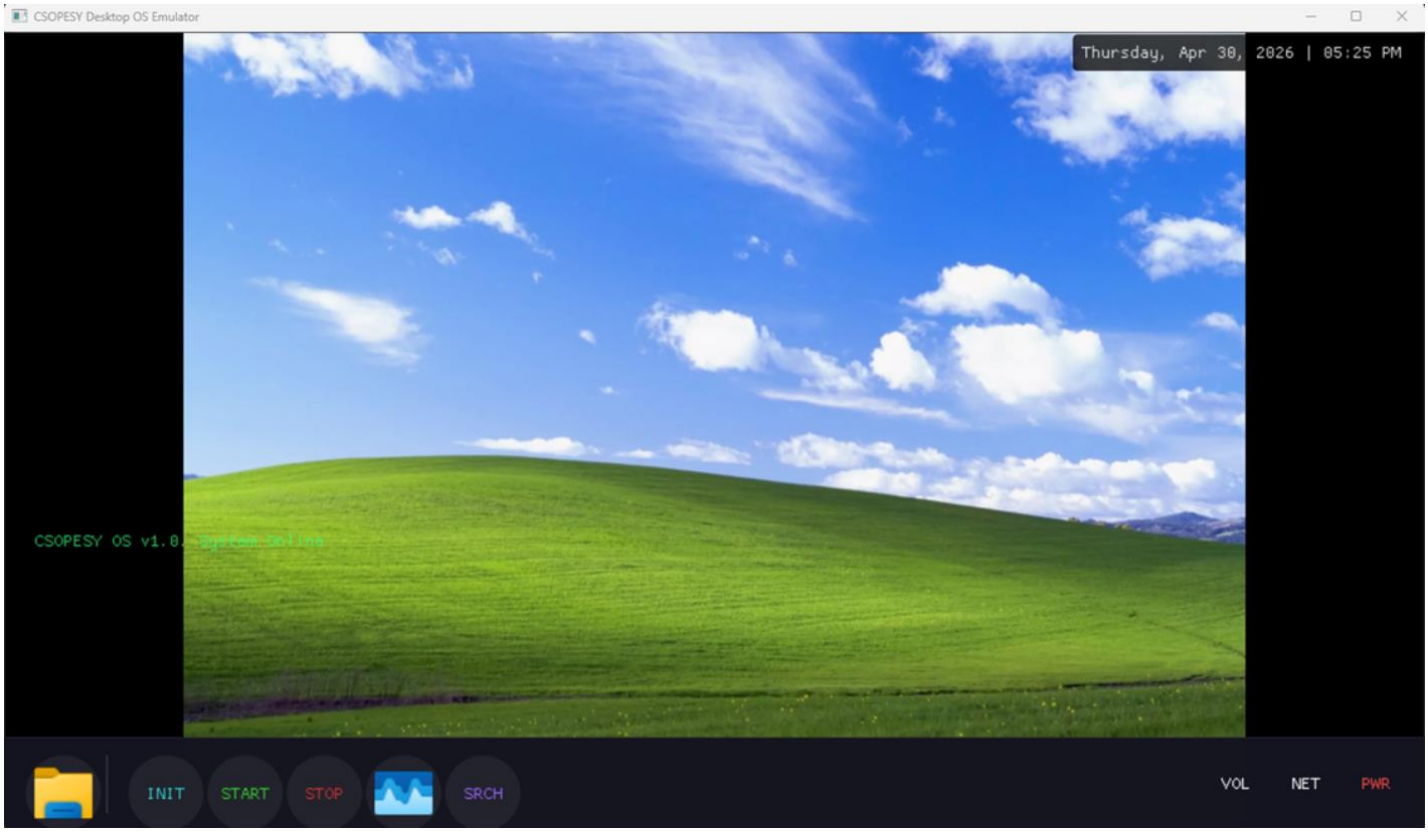
Updated as of: Sept 18, 2026

**General Instructions:** Desktop-Style OS Mock-up

Implement your own desktop-style OS mockup, using dear IMGUI - an immediate-mode UI library. Just as Windows, macOS, and Android have a compositor that manages windows and renders the desktop every frame, your CSOPESY emulator will have its own compositor built on top of the GLFW + OpenGL + Dear ImGui stack. The mockup must simulate the visual experience of a minimal operating system shell — not a fully functional OS, but a convincing, real-time graphical environment that demonstrates mastery of immediate-mode UI design.






**Checklist of Requirements**
Your system must have ALL the following features implemented properly.

| **Requirement** | Deskop |
|---|---|
| **Description** | A full-screen background that serves as the base layer of the OS. This is the first thing rendered each frame.<br>• Fills the entire application window<br>• Displays a **wallpaper** — either a solid color gradient, a pattern drawn using ImGui draw commands, or a loaded image texture<br>• Displays a **real-time clock** (current time, updated every frame) in a fixed corner of the screen<br>• Provide a PWR button that serves as a shutdown. The application must be closed using this button, and not by force exit. |
| **Requirement** | Taskbar |
| **Description** | A fixed panel (top or bottom of screen) that provides access to system functions and shows running applications.:<br>• "Contain at least three clickable icon buttons. It's up to you how to design these buttons. Two of these buttons must open up a **unique** UI screen, with placeholder information. Any UI screen with your own design or layout is acceptable. The third button must serve as the **Task Manager.** |
| **Requirement** | Task Manager |
| **Description** | A window that closely resembles the Windows task manager:<br>• Must have a placeholder table showing "Processes" and their respective CPU and memory usage<br>• Use dummy values on the Task Manager |

**ASSESSMENT METHOD**
The project will be assessed through a black box quiz system in a time-pressure format. This is to minimize drastic changes or "hacking" your project to ensure the test cases are met. You should only modify the parameters and no longer recompile the project when taking the quiz.

Test cases, parameters, and instructions are provided per question, wherein you must submit a video file (.MP4), demonstrating your project. Some questions will require submitting PowerPoint presentations, such as cases explaining the details of your implementation.

**IMPORTANT DATES**
See AnimoSpace for specific dates.

| | |
|---|---|
| **Week 5** | Project submission |

**Submission Details**
Aside from video files for the quiz, you need to prepare some of the requirements in advance, such as:

- SOURCE - Contains your source code. Add a README.txt with your name and instructions on running your program. Also, indicate the entry class file where the main function is located. An alternative can be a GitHub link.
- PPT – A technical report of your system containing:
    - Cover
    - Video Walkthrough
    - Architectural Diagram
    - Code Snippets
    - Provide a written discussion on how you designed your desktop-style OS
- Video
    - Your video must be seamless and not cut. The minimum video quality should be at least 480p and a maximum of 720p. Please ensure that the font size of your IDE is large enough to be readable on video.
    - Maximum video size should be limited to 1GB.
    - You are not allowed to access/modify your code while performing the test case.
    - The video should always show you pressing Run/Debug from your IDE, which should initialize your program. Once running, you are not allowed to exit from your program and attempt to modify your code.
    - Not following upload instructions will invalidate your submission.
    - Ensure that your MP4 is directly embedded in your PPTX. It should be reflected in your total file size

**Grouping**
This is a group project. Each group can have a maximum of four members. All members of the group should be in the same section.

**Grading Scheme**

- You are to provide evidence for each test case, recorded through video. Each test case will have some points allocated. The test cases will be graded as follows:

| **Functional** | | |
|---|---|---|
| No points | Partial points | Full points |
| The desktop emulator did not pass the test case. **NO WORKAROUND** is available to produce the expected output. | The desktop emulator did not pass the test case. **A workaround** is available to produce the expected output. | The desktop emulator passed the test case using varying inputs and produced the expected output. |